

Monci Mamachan

✉ moncimamachan8547@gmail.com

📍 Dublin, Ireland

🌐 linkedin.com/in/monci-mamachan

☎ +353 899642672

🔗 <https://monci-mamachan628.github.io/portfolio/>

🐙 <https://github.com/monci-mamachan628>

Professional Summary

Detail-oriented and results-driven Data Analyst with a Master's in Data Science for Management. Proficient in Python, R, SQL, and Power BI, with hands-on experience in data cleaning, transformation, and statistical analysis. Skilled in delivering actionable insights and building impactful dashboards to support data-driven decision-making.

WORK EXPERIENCE

10/2024 – 12/2024
Catania, Italy

Data Analyst Intern
MDB Financial Consultancy

- Migrated local projects to GitHub for version control and team collaboration.
- Set up CI/CD pipelines for automated integration and deployment.
- Automated deployments to Firebase and Google Cloud for seamless production updates.
- Improved deployment efficiency by reducing manual processes through automation.
- Configured Firebase Security Rules for enhanced application security.

12/2022 – 06/2023
Catania, Italy

Data Analyst Intern
ST Microelectronics

- Developed application software for video magnification using MATLAB, enhancing visibility of subtle motion in visual data.
- Learning motion magnification theory and its application in Artificial Intelligence, to implement it in laboratory settings for predictive reinforcement learning and reliability maintenance of power devices.
- Analyzed technical challenges and collaborated with cross-functional teams to design effective solutions.
- Provided project progress updates and proposed solutions to issues.

Education

10/2020 – 12/2023
Catania, Italy

MASTERS IN DATA SCIENCE FOR MANAGEMENT
University of Catania

Field of study: Data Science for Management
Thesis: Learning Motion Magnification Theory/Application

2015 – 2019
Kerala, India

BACHELOR OF TECHNOLOGY ENGINEERING - COMPUTER SCIENCE
APJ Abdul Kalam Technological University

Project: College Enquiry Chatbot

SKILLS

Tools & Technologies

- Microsoft Excel
- Tableau
- Power BI
- MySQL
- Matlab

Programming

- Python (NumPy, Pandas, Seaborn, Matplotlib)
- R
- C
- JavaScript
- Tailwind CSS

PROJECTS

Medicare Drug Prescriptions & Cost Analysis, 2023 [↗](#)

This project presents an intermediate-level data analytics case study using the 2023 Medicare Part D Prescriber Public Use File (PUF). It aims to explore prescription costs, drug classes, demographic disparities, and public health trends in the U.S.

Technical Skills: Python, Pandas, ETL, Power BI, AdHoc Reporting

Implement a deep-learning pipeline for classification of pneumonia in X-Ray images. [↗](#)

Designed and implemented a robust deep learning pipeline for the detection and classification of pneumonia from chest X-ray images. Key aspects include data preprocessing and augmentation techniques, such as SMOTE for class balancing, as well as fine-tuning of state-of-the-art deep learning models (VGG-16 and ResNet-50) to achieve strong performance on real-world medical imaging data.

Technical Skills: Python libraries like Pandas, NumPy and ETL data analysis(EDA), Hypothesis testing, Logistic regression analysis, correlation studies.

Renewable Energy Dashboard [Power BI | Excel]

Developed an interactive Power BI dashboard for wind and solar plants, leveraging advanced DAX measures and dynamic visualizations to analyze key performance indicators (KPIs) such as capacity factor, plant availability, energy yield with statistical analysis. Implemented detailed drill through reports and heatmaps to provide in-depth insights into the correlation between variables like wind speed, solar irradiance, and active power generation, enabling stakeholders to make data-driven decisions and optimize plant operations.

HR Data Analytics- HR Domain [Power BI | Excel]

Designed a Power BI dashboard to track employee data for the HR team, including working hours, attendance, performance, and leaves. The dashboard streamlined HR processes and increased efficiency. This dashboard can save 3-hrs work for the HR daily.

Enhancing Car Loan Approvals: Leveraging Random Forests for Risk Prediction

Utilized a random forest algorithm to enhance the accuracy of car loan approvals and minimize default risk. The project involved comprehensive data preprocessing, model development, performance assessment using financial indicators, and hyperparameter optimization, demonstrating the real-world value of machine learning in financial decision-making.

Technical Skills: Machine Learning, Random Forest, Financial Risk Analysis, Model Optimization

Twitter Sentimental Analysis

Used Machine Learning and Natural Language Processing to identify the sentiment of tweets dataset.

Technical Skills: Natural Language Processing (NLP), Embedding.

Data Visualization using TABLEAU and Google Data studio

Performed data visualization and analysis using Tableau to identify trends, patterns, and insights. Also analyzed the World Happiness Index dataset using Google Data Studio for interactive dashboards and reports.

Technical skills: Google Data Studio, Big query

Certificates

- Advanced SQL Certification (Hacker Rank) [↗](#)
- SQL for Data Science with Python (IBM) [↗](#)
- Prompt Engineering for Everyone (Cognitive Class) [↗](#)
- Data Analysis with Python [↗](#)